

Non-Terrestrial Networks for 5G and 6G: A Survey of 3GPP Standardization and an Application-Driven Architecture Selection Framework

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Abstract—Non-Terrestrial Networks (NTNs) are now a formal part of 5G New Radio and a foundational component of the 6G vision for ubiquitous connectivity. Over the last few releases, 3GPP has turned NTN from a research idea into something that can actually be built. Release 17 standardized satellite access for NR, and Releases 18 and 19 are extending the work toward regenerative payloads, Ka-band operation, and direct-to-device service. What is still missing is a clear answer to a more practical question: which NTN architecture fits which deployment scenario. This paper surveys NTN research and standardization across architecture and topology, physical layer challenges including Integrated Sensing and Communication (ISAC), mobility and handover, network slicing, security and resilience, and application-specific deployments. We also discuss emerging 6G enablers such as AI-driven beamforming, reconfigurable intelligent surfaces, optical inter-satellite links, orbital angular momentum, and quantum key distribution. Building on the survey, we propose an application-driven mapping framework that recommends a concrete 3GPP configuration for a given deployment. We formalize the framework as a multi-criteria decision-making (MCDM) problem with a weighted sum model extended with sensing and environmental impact criteria, and outline fuzzy and Bayesian extensions for regulatory inputs and empirical updating. The headline finding is that no single architecture wins everywhere: LEO with a regenerative payload is the most deployment-ready option for rural broadband today, UAV and HAPS suit disaster recovery, and for maritime and aviation the binding constraint is regulatory rather than physical.

Index Terms—Non-Terrestrial Networks, 5G NR, 6G, LEO, UAV, HAPS, 3GPP, RIS, AI, network slicing, MCDM, NR-NTN, ISAC, QKD

I. INTRODUCTION

Even after several decades of cellular densification, large parts of the world still have either no coverage or coverage that is too unreliable for modern services. Remote rural regions, oceans, conflict zones, and post-disaster areas are the most obvious examples. Terrestrial networks are limited by geography and by the economics of laying fiber and building towers, and no amount of small-cell densification fixes a problem that is fundamentally about reach. Non-Terrestrial Networks (NTNs) address this gap by adding satellites, high-altitude platform

stations (HAPS), and unmanned aerial vehicles (UAVs) to the cellular architecture as either airborne base stations or as relay nodes [1].

Two trends have made this a particularly active area in the last few years. First, large LEO constellations such as Starlink, OneWeb, and Amazon Kuiper have moved from concept to commercial deployment, demonstrating that broadband from space is feasible at meaningful scale [4]. Second, 3GPP introduced NTN into the 5G specification in Release 17, giving the industry a common standard rather than a collection of proprietary systems [3], [8]. Release 18 (5G-Advanced) extended the work into Ka-band and added support for vehicle-mounted earth stations, and Release 19 is moving toward regenerative payloads, store-and-forward IoT, and HAPS as active gNBs [5]. In parallel, the 6G research community has begun to fold NTN into a broader vision that includes AI-native networks, reconfigurable intelligent surfaces (RIS), terahertz feeder links, ISAC, and post-quantum security [10].

3GPP has supplied a useful toolbox of mechanisms. What it has not supplied, and what existing surveys have not really captured either, is guidance on which combination of these mechanisms makes sense for which application. Prior works such as [1], [4], [10] provide thorough treatments of NTN mechanisms but none formalizes an application-to-architecture mapping, and none incorporates sensing capability or environmental sustainability as explicit selection criteria. That is the gap this paper tries to close. We survey the standardization status across Releases 17 to 19, compare the main NTN architectures along several axes, and propose a framework that maps an application requirement set to a recommended 3GPP configuration. The framework is formalized as a multi-criteria decision-making (MCDM) problem so the reasoning is explicit and traceable. The rest of the paper is organized as follows. Section II reviews NTN taxonomy and 3GPP standardization. Section III presents the comparative survey. Section IV formalizes the mapping framework. Section V discusses results and a sensitivity analysis. Section VI concludes.

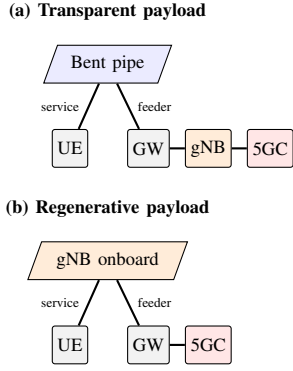


Fig. 1. NTN payload architectures. The transparent design (a) puts the gNB on the ground and uses the satellite as a bent pipe; the regenerative design (b) hosts the gNB onboard and removes the ground-gateway double hop.

II. BACKGROUND AND 3GPP STANDARDIZATION

A. NTN Taxonomy and Payload Architectures

The NTN family contains three main groups of platforms: spaceborne satellites (LEO at roughly 500–1200 km, MEO at 8000–20000 km, GEO at 35786 km), HAPS at around 20 km altitude, and lower-altitude UAVs operating below a few hundred meters [1], [4]. Each has a different orbital or flight dynamic, propagation profile, and cost structure, which is part of why a single set of design assumptions cannot cover all of them.

A second axis that matters for system design is whether the satellite payload is *transparent* or *regenerative*. As shown in Fig. 1, a transparent payload acts as a frequency-shifting bent pipe, with the gNB sitting on the ground behind the feeder link. A regenerative payload puts the gNB, or at least its distributed unit, onboard the satellite [6]. Release 17 standardized the transparent case first because it is simpler and matches what most early commercial systems do. Regenerative payloads are part of the Release 19 roadmap and reduce end-to-end latency by removing the double hop through the ground gateway. That said, as of early 2026 roughly two-thirds of active LEO broadband payloads still rely on transparent architectures, largely for lower on-orbit complexity and faster time-to-market [4]. Regenerative designs remain the better choice for latency-sensitive workloads, but transparent payloads continue to dominate commercial baseline broadband deployments.

B. 3GPP Releases 15 through 19

The 3GPP work on NR-NTN started in Release 15 with a study of channel models in TR 38.811 [7], which produced NTN-specific channel models accounting for free-space path loss, clutter and shadow fading at different elevation angles, and atmospheric and scintillation losses at S-band and Ka-band. Release 16 followed with TR 22.822 [9], defining satellite access use cases and service requirements, and TR 38.821 [8], identifying necessary changes at the architecture, higher-layer, and physical layers.

Release 17 turned the study into a normative work item, and several of its mechanisms are now central to NTN deploy-

ments. The user equipment (UE) is assumed to have GNSS capability, which it uses with broadcast satellite ephemeris to pre-compensate for both Doppler shift and propagation delay before transmitting on the uplink [3]. The HARQ process count was extended from 16 to 32, and HARQ feedback can be disabled per process to avoid stalling on long round trips. The network signals a common timing advance (TA) for the satellite-to-gNB round trip, and the UE adds its own UE-to-satellite component. Earth-fixed tracking areas were introduced so moving cells do not cause endless tracking-area updates. Conditional handover was extended with triggers based on UE location and the timing of satellite coverage. New radio access types NR(LEO), NR(MEO), and NR(GEO) were added so the core network and policy functions can distinguish satellite types, and a separate work item added NB-IoT and LTE-M support over NTN.

Release 18 (5G-Advanced, 2024) extended NR-NTN above 10 GHz, added vehicle-mounted earth station (VMES) and earth station on vessel (ESV) mobility, and improved IoT-NTN handling of discontinuous coverage. Release 19 continues with regenerative payload support, store-and-forward operation for extreme-remote IoT, and explicit HAPS and UAV roles as active gNBs. The Release 20 study item on ISAC (TR 38.914) is the latest addition, examining how NTN nodes can concurrently transmit communication waveforms and extract sensing measurements from the same spectral resource [17], [18].

III. COMPARATIVE SURVEY

A. Architecture and Network Topology

Table I summarizes the main NTN platform types along altitude, latency, cost, mobility, and typical frequency bands. LEO is currently the most active segment, both because of the existing constellations and because its latency profile is good enough for interactive services. GEO offers very wide coverage from a single satellite but the round-trip delay rules it out for any latency-sensitive workload. HAPS sits in an interesting middle ground, with regional coverage and low-millisecond latency, though the platforms themselves are not yet at scale. UAVs are very flexible but limited by energy and operational range, which is why they tend to come up for short-duration deployments rather than persistent coverage. A subtle point is that the latency in Table I is the link budget alone; the end-to-end user latency in a transparent-payload deployment also includes the feeder link.

B. Physical Layer Challenges

Three classical issues dominate the NR-NTN physical layer. The first is Doppler: a LEO satellite at 7.5 km/s induces shifts of roughly 24 kHz at 2 GHz and 240 kHz at 20 GHz, far above terrestrial NR receiver design [3]. The 3GPP solution pushes Doppler pre-compensation onto the UE using GNSS position and ephemeris. The second is path loss: multipath is weak because the long satellite distance collapses the angular spread to near zero, leaving free-space loss dominant with corrections for atmospheric absorption, scintillation, and elevation-dependent shadow fading [7]. The third is timing:

TABLE I
COMPARISON OF NTN PLATFORM TYPES

Type	Altitude	Latency	Cost	Mobility	Bands
GEO	35786 km	~135 ms	High	Earth-sync.	C, Ku, Ka; L/S
MEO	8–20k km	40–80 ms	High	Hours-scale	L, S, C
LEO	0.5–1.2k km	1.5–10 ms	Med.	7.5 km/s	Ku, Ka; L/S
HAPS	~20 km	~0.1 ms	Med.	Quasi-static	S (IMT), mmW
UAV	<1 km	<0.05 ms	Low	Highly dyn.	Sub-6 GHz

the long round trip breaks the assumption that uplink timing closely follows downlink timing, which the Release 17 split TA addresses directly.

Several emerging techniques aimed at 6G are worth flagging. AI-driven beamforming is an active area: predicting where a moving LEO beam should point a few hundred milliseconds ahead is essentially a regression problem, and learned models trained on ephemeris and historical link data can outperform analytic predictors when the channel has structure the analytic model misses [12]. Reconfigurable intelligent surfaces (RIS) place a passive surface of tunable elements between transmitter and receiver to shape the wireless environment electronically [11]; mounting an RIS payload on a HAPS or UAV creates a low-energy controllable relay that fills coverage gaps or compensates for high free-space loss in Ka-band. For inter-satellite links (ISLs), optical links offer multi-Gbps rates with narrow beams that resist interception, while RF ISLs are slower but more robust; most current LEO constellations use optical ISLs for the constellation backbone. For 6G point-to-point feeder links, orbital angular momentum (OAM) multiplexing adds spectrally efficient channels by exploiting mode orthogonality.

Integrated Sensing and Communication (ISAC). A growing area within the Release 20 study item TR 38.914 is the dual use of NTN communication waveforms for environmental sensing [17], [18]. Satellites and HAPS occupy vantage points unavailable to ground infrastructure, and the same signal used for broadband access can, with appropriate receiver processing, yield radar-grade measurements. Applications include sea-state monitoring from LEO via Doppler shift in reflected signals, space debris tracking, and surface change detection for disaster assessment. HAPS are particularly attractive because their quasi-stationary position allows coherent radar integration over much longer dwell times than a fast-moving LEO pass allows. The practical challenge is that aggressive communication beamforming conflicts with the wide illumination patterns needed for sensing, so current research focuses on joint waveform design and time-division ISAC frames.

C. Mobility and Handover

Mobility in NTN behaves differently from anything in terrestrial cellular. In a LEO constellation with moving beams,

the cell footprint sweeps across the ground at orbital speed, so even a stationary user experiences handovers every few minutes. The 3GPP response was Earth-fixed tracking areas, which decouple the geographic identity of a tracking area from the satellite illuminating it, and conditional handover with location-based and coverage-window-based triggers [3]. UAV mobility is a different problem: UAVs can be repositioned quickly but they fight for spectrum with terrestrial base stations and introduce strong air-to-ground line-of-sight components, which raises interference into neighboring cells [2].

Machine learning is beginning to make its way into handover decisions for NTN. The standardized conditional-handover triggers are deterministic rules that fire when measurements cross thresholds. A learned predictor can use the same ephemeris information plus user trajectory and historical link quality to anticipate handovers earlier and reduce ping-pong, which matters when the natural cadence of LEO handovers is already a few per session [12].

D. Network Slicing in NTN

Network slicing extends naturally from terrestrial 5G to NTN, but the wide-area, time-varying nature of satellite links makes resource isolation between slices harder than in fixed networks. The standardized slice types (eMBB, URLLC, mMTC) carry very different requirements, and on a shared LEO beam they compete for the same time-frequency resources [13]. The current 3GPP approach extends slice management to NTN-aware orchestration so that a low-latency emergency slice is not throttled by a high-bandwidth video slice sharing the same satellite. A practical complication is that per-slice SLAs may need relaxation during low-elevation passes when the link budget is worst, and the standard does not yet define how that renegotiation is exposed to the application layer.

E. Security and Resilience

NTN inherits the 5G security architecture but introduces threats that terrestrial networks do not face. Jamming and spoofing of satellite signals are easier than for terrestrial cells because the wide footprint makes a single ground-based jammer effective over a large area, and GNSS spoofing in particular can disrupt the GNSS-based timing pre-compensation that NR-NTN relies on [14]. Mitigations include frequency hopping in higher bands, cross-link authentication via constellation-internal routes, and beamforming nulls toward identified jammer locations.

For inter-satellite links, quantum key distribution (QKD) is emerging as a longer-term option for securing the constellation backbone. Ground-to-satellite QKD experiments, including the Micius satellite programme, have demonstrated single-photon key exchange over distances exceeding 1000 km [19]. Integrating QKD into the ISL layer would provide information-theoretic security against adversaries with unlimited computational resources, including future quantum computers, but practical barriers remain: pointing precision, atmospheric turbulence on the uplink, and the absence of a

TABLE II
APPLICATION TO ARCHITECTURE MAPPING

Use Case	Best Architecture	Key 3GPP Mechanism
Rural broadband	LEO regenerative	Earth-fixed TA, conditional HO, eMBB slice
Disaster recovery	UAV / HAPS gNB	IoT-NTN (NB-IoT, LTE-M), URLLC slice
Maritime, aviation	LEO with optical ISL	Extra-territorial PLMN, VMES/ESV mobility

3GPP framework for quantum-derived key material alongside classical 5G credential stores.

A related concern is data sovereignty: when a regenerative payload processes traffic in orbit across multiple jurisdictions, end-to-end encryption alone does not determine which country’s lawful-intercept rules apply, which is a regulatory question Release 18 SA1 flags but does not resolve [3].

F. Application-Specific Deployments

Table II captures the headline application-to-architecture mapping. The key insight is that the right combination of architecture, payload type, and 3GPP mechanism depends heavily on what the deployment is for.

Rural broadband. For fixed and quasi-fixed users in low-density regions, LEO is currently the most viable option. Starlink has shown the model works at scale: as of late 2024 it had on the order of four million subscribers across more than 100 countries [16], with measured median downlink throughput around 100 Mbps and one-way latency typically below 50 ms [15]. A large fraction of that fleet still uses transparent payloads, which is a useful reminder that regenerative designs are the better architecture on paper but are not a prerequisite for commercial viability at scale today. Earth-fixed tracking areas keep the core network from being flooded with TA updates as satellites move, and conditional handover smooths the user experience across beam switches. The remaining bottleneck is economic, not technical. Phased-array user terminals still cost \$300–600, which is high relative to a typical rural household budget, and is why NB-IoT-NTN is more interesting than full broadband for many rural IoT and agricultural use cases [3], [4].

Beyond fixed-terminal broadband, the most commercially active recent development is direct-to-device (D2D) service from LEO, where an unmodified smartphone connects directly to a satellite without specialized hardware [21]. Starlink Direct to Cell and AST SpaceMobile’s BlueBird have demonstrated LTE-handset connectivity in field trials at narrowband IoT throughputs today, with 3GPP TR 38.821 defining the capability classes for a trajectory toward higher data rates and broader rural coverage.

Disaster recovery. When terrestrial infrastructure is lost or degraded after a flood, earthquake, or wildfire, what matters is how quickly coverage can be re-established over a specific impact zone. A LEO constellation has wide coverage but

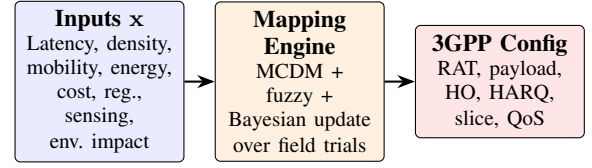


Fig. 2. Application-driven mapping framework. Deployment requirements are translated into a concrete 3GPP-level NTN configuration.

cannot reconfigure its footprint to focus on a single town within hours. UAV-mounted gNBs and HAPS can. UAVs can be flown to the area within a day, hover for hours, and provide a few square kilometers of coverage with low-latency air-to-ground links [2]. HAPS extend this to regional scale at low double-digit milliseconds. The standardization work supporting this case is mostly on the IoT-NTN side, where NB-IoT and LTE-M over NTN allow low-power devices, including survivor-locator beacons, to report position and status when terrestrial cell sites are down. Emergency response trials in Japan and Italy have demonstrated that a UAV gNB can come online in under an hour, well inside the window where it materially affects rescue operations [20].

Maritime and aviation. For ships in international waters and aircraft over oceans, terrestrial coverage simply does not exist. LEO with optical inter-satellite links is the natural fit because ISLs let traffic move between satellites without requiring a ground station within radio range, which is critical over the Pacific or polar regions. The technical mechanisms are mature; the binding constraint, somewhat unusually, is regulatory. Satellite beams cover multiple countries at once and legacy public land mobile network (PLMN) selection assumes country-bounded networks. Release 18 SA1 work on extra-territorial PLMN selection addresses this directly [3], and VMES/ESV mobility provides the radio-level basis for vessels and aircraft to operate as mobile UEs.

IV. APPLICATION-DRIVEN ARCHITECTURE SELECTION

Existing surveys describe NTN mechanisms but stop short of recommending which to use for what. We propose a small framework, illustrated in Fig. 2, that closes this gap. It takes a vector of application requirements, runs them through a mapping engine seeded with surveyed evidence and field-trial data, and emits a concrete 3GPP-level configuration: RAT type (NR-LEO, NR-MEO, NR-GEO), payload choice, handover strategy, HARQ configuration, slice type, and 5G QoS class.

A. Input Vector and Weights

We represent application requirements as a normalized vector

$$\mathbf{x} = [x_{\text{lat}}, x_{\text{dens}}, x_{\text{mob}}, x_{\text{pwr}}, x_{\text{cost}}, x_{\text{reg}}, x_{\text{sens}}, x_{\text{env}}]^T, \quad (1)$$

with each $x_i \in [0, 1]$ encoding sensitivity to that requirement (1 = very sensitive, 0 = tolerant). Two additions relative to the baseline framework are worth noting. The sensing sensitivity x_{sens} captures whether the deployment expects the NTN node to concurrently perform ISAC functions such as sea-state

monitoring or debris detection; applications that are purely communication-oriented set $x_{\text{sens}} = 0$. The environmental impact sensitivity x_{env} reflects sustainability constraints: a high value penalizes architectures that require large-scale constellation launches when a lower-footprint HAPS or UAV solution meets the minimum service requirements. This is increasingly relevant given growing regulatory scrutiny around orbital debris and spectrum coexistence [10]. A weight vector $\mathbf{w}$ with $\sum_i w_i = 1$ captures the relative importance of each requirement. For rural broadband, plausible weights are $w_{\text{cost}} = 0.35$, $w_{\text{lat}} = 0.25$, $w_{\text{reg}} = 0.15$, $w_{\text{dens}} = 0.10$, $w_{\text{mob}} = 0.05$, $w_{\text{pwr}} = 0.05$, $w_{\text{sens}} = 0.0$, $w_{\text{env}} = 0.05$.

B. Performance Matrix and Decision Rule

A performance matrix $\mathbf{P}$ has, for each candidate architecture A_j and each requirement i , a score $p_{j,i} \in [0, 1]$ indicating how well that architecture meets that requirement. Scores are derived from link-budget analysis and published field-trial data, and are subject to Bayesian update as described in Section IV-C. Table III gives an indicative example, now including a sensing column (p_{sens}) that reflects each platform's ability to support concurrent ISAC functions. HAPS scores highest on sensing because its quasi-stationary geometry enables coherent radar integration over extended dwell times; LEO scores well for wide-area sensing passes; GEO is useful for large-scale change detection but has limited spatial resolution at that altitude. Environmental impact (p_{env} , not shown in the table for width reasons) would penalize LEO most heavily due to constellation scale, while UAV and HAPS fare better on this criterion. The mapping engine computes a weighted sum

$$S_j = \sum_{i=1}^n w_i p_{j,i}, \quad (2)$$

for each architecture. WSM was chosen over rank-based alternatives such as TOPSIS and VIKOR for its transparency: S_j is a direct linear combination of weights and scores, so each recommendation is traceable to individual criteria without rank-reversal ambiguity. The unconstrained selector is

$$A^* = \arg \max_j S_j. \quad (3)$$

Selection is restricted, however, to the feasibility set

$$\mathcal{F} = \{A_j : p_{j,i} \geq x_i \quad \forall i \in \mathcal{H}\}, \quad (4)$$

where $\mathcal{H} \subseteq \{1, \dots, n\}$ collects hard-constraint criteria (density is a member of $\mathcal{H}$ in the rural broadband case). The final decision is $A^* = \arg \max_{j \in \mathcal{F}} S_j$.

For the rural broadband weights above, applying (2) yields $S_{\text{LEO}} \approx 0.62$, $S_{\text{GEO}} \approx 0.55$, $S_{\text{HAPS}} \approx 0.66$, and $S_{\text{UAV}} \approx 0.71$. The naive argmax picks UAV, but UAV scores 0.20 on density and is eliminated by (4) because its $p_{\text{UAV,dens}} < x_{\text{dens}}$ for any non-trivial coverage requirement. After applying the feasibility filter, LEO is selected. For extremely remote scenarios such as a single research station or a small expedition camp, x_{dens} is itself low, the constraint is non-binding, and the framework legitimately returns UAV or HAPS as the winner rather

TABLE III
INDICATIVE PERFORMANCE MATRIX $\mathbf{P}$

Arch.	p_{lat}	p_{dens}	p_{mob}	p_{pwr}	p_{cost}	p_{reg}	p_{sens}
GEO	0.10	0.90	0.30	0.40	0.80	0.30	0.55
MEO	0.40	0.80	0.40	0.40	0.60	0.40	0.50
LEO	0.85	0.70	0.50	0.55	0.55	0.55	0.75
HAPS	0.95	0.50	0.80	0.65	0.65	0.70	0.85
UAV	1.00	0.20	0.95	0.30	0.90	0.85	0.60

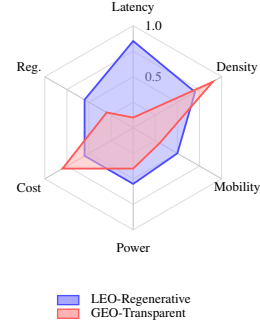


Fig. 3. Radar chart comparing LEO-Regenerative and GEO-Transparent architectures across six of the eight MCDM criteria (sensing and environmental impact omitted for visual clarity). LEO dominates on latency; GEO is stronger on density coverage and cost per bit for broadcast workloads.

than LEO. The framework then emits the matching 3GPP configuration: regenerative payload, conditional handover with coverage-window triggers, HARQ extended to 32 processes with feedback enabled, and an eMBB slice with standard 5G QoS classes.

Fig. 3 visualizes the trade-off between LEO-Regenerative and GEO-Transparent across the six core criteria. The radar chart illustrates why GEO is competitive on density and cost yet is disqualified for any latency-sensitive workload, and how LEO's advantage on latency and sensing comes at the expense of regulatory and cost scores relative to GEO.

C. Fuzzy and Bayesian Extensions

Two extensions improve the framework. First, the regulatory input is rarely a clean scalar. A fuzzy inference system can handle inputs like “international waters,” “single country,” and “conflict zone” through rules of the form: if regulatory scope is high complexity AND ISL is available, then confidence in LEO is high. Fuzzification keeps the math in (2) unchanged but lets x_{reg} be derived from richer inputs. Second, the matrix $\mathbf{P}$ should not be static. Modelling each score as the success probability in a Bernoulli process, a Beta-Binomial conjugate update gives a closed-form posterior mean. Given prior hyperparameters (α_0, β_0) and N field observations of which k confirm the prior score:

$$p'_{j,i} = \frac{\alpha_0 + k}{\alpha_0 + \beta_0 + N}. \quad (5)$$

The prior is seeded from link-budget analysis and published performance surveys; the posterior is shaped by Starlink

FCC submissions, IoT-NTN trials, and post-mortems of failed programmes such as Loon.

As a concrete illustration: the prior LEO latency score $p_{\text{LEO,lat}} = 0.80$ corresponds to $(\alpha_0, \beta_0) = (4, 1)$ (prior mean $\alpha_0/(\alpha_0 + \beta_0) = 4/5 = 0.80$). After $N = 20$ 2025 Starlink field measurements with $k = 17$ confirming median one-way latency below 40 ms, (5) yields $p'_{\text{LEO,lat}} = 21/25 = 0.84$. The update is modest but in close decisions between LEO and HAPS it can be decisive. The empirical layer is what keeps the framework from drifting away from what actually deploys.

V. COMPARATIVE ANALYSIS AND DISCUSSION

Across the survey, the ranking by deployment readiness for rural broadband is clear. LEO with a regenerative payload, conditional handover, and the Release 17 timing-advance and HARQ extensions is the most viable option today. Transparent payloads remain the commercial baseline for most of the fleet, which is a practical reminder that the theory-versus-deployment gap in NTN is real and worth tracking. GEO retains a niche for broadcast and backup coverage where latency does not matter. HAPS and UAV are technically attractive for localized and burst-coverage scenarios, but their standardization as active gNBs sits on the Release 19 timeline rather than being a finished product.

Sensitivity analysis. Table IV quantifies S_j under three perturbations to the base rural broadband case. For the environmental scenario, assumed p_{env} values are: LEO=0.20 (constellation scale), HAPS=0.85, GEO=0.70; the weight shift is $w_{\text{env}}: 0.05 \rightarrow 0.15$, $w_{\text{cost}}: 0.35 \rightarrow 0.25$. UAV is excluded by the density feasibility constraint in all scenarios.

TABLE IV
SENSITIVITY OF S_j TO PARAMETER PERTURBATIONS (RURAL BROADBAND)

Scenario	S_{LEO}	S_{GEO}	S_{HAPS}	Winner
Base (rural broadband)	0.62	0.55	0.66	LEO
Terminal cost -50%	0.71	0.55	0.66	LEO
Reg. complexity $\uparrow$	0.64	0.53	0.66	LEO
High w_{env} (0.15)	0.59	0.54	0.68	HAPS [†]

[†]Small-area deployments only; LEO preferred at regional scale.

Open issues. Two gaps remain. First, LEO terminal cost (\$300–600) is still prohibitive in many markets; the hardware roadmap matters as much as the air-interface one. Second, regulatory and security questions : extra-territorial coverage, GNSS spoofing, data sovereignty, and quantum-safe key management, are likely the slowest-moving pieces.

VI. CONCLUSION AND FUTURE WORK

This paper surveyed NTN standardization across Releases 17–19 and proposed an application-driven MCDM framework mapping deployment requirements to concrete 3GPP configurations, extended with ISAC sensing and environmental sustainability criteria, fuzzy regulatory inputs, and Bayesian field-trial updating. Four observations stand out: 3GPP has provided a usable toolbox; no single architecture wins everywhere; transparent payloads dominate commercial deployments even

as regenerative designs offer better latency on paper; and sustainability and ISAC are emerging as first-class 6G design criteria. Open directions include direct-to-device LEO service, HAPS and UAV as active gNBs, learned handover models, joint ISAC waveform design, and field-trial validation of the framework.

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